Spanning Connectivity of the WK-Recursive Networks

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Abstract

A k-container $C(u, v)$ of G between u and v is a set of k internally disjoint paths between u and v. A k-container $C(u, v)$ of G is a k^*-container if it contains all vertices of G. A graph G is k^*-connected if there exists a k^*-container between any two distinct vertices. The spanning connectivity of G, $\kappa^*(G)$, is defined to be the largest integer k such that G is w^*-connected for all $1 \le w \le k$ if G is an 1^*-connected graph and undefined if otherwise. We use $K(d, t)$ to denote the WK-recursive network of level t, each of which basic modules is a d-vertex complete graph, where $d > 1$ and $t \ge 1$. In this paper, we prove that the $\kappa^*(K(d, t)) = d-1$ for $d \ge 4$ and $t \ge 1$.

1 Introduction

For the graph definitions and notations we follow [2]. $G = (V, E)$ is a *graph* if V is a finite set and E is a subset of $\{(u, v) \mid (u, v)$ is an unordered pair of $V\}$. We say that V is the *vertex set* and E is the *edge set*. A graph H is a *subgraph* of graph G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. Two vertices u and v are *adjacent* if (u, v) is an edge of G. The set of *neighbors* of u, $N_G(u)$, is $\{v \mid (u, v) \in E\}$. The *degree* of a vertex u of G, $\deg_G(u)$, is the number of edges incident with u. A graph G is *k-regular* if $\deg_G(u) = k$ for every vertex u in G. A *path* is a sequence of vertices represented by $\langle v_0, v_1, \ldots, v_k\rangle$ with no repeated vertices and (v_i, v_{i+1}) is an edge of G for all $0 \le i \le k-1$ when $k \ge 1$ and $\langle v_0\rangle$ when $k = 0$. We also write the path $P = \langle v_0, v_1, \ldots, v_k\rangle$ as $\langle v_0, \ldots, v_i, Q, v_j, \ldots, v_k\rangle$, where Q is a path from v_i to v_j. We use P^{-1} to denote the path $\langle v_k, v_{k-1}, \ldots, v_0\rangle$. The *length* of a path Q, $l(Q)$, is the number of edges in Q. The *distance* of vertices u and v of G, $d_G(u, v)$, is the length of the shortest path between u and v. A path is a *hamiltonian path* if it contains all vertices of G. A graph G is *hamiltonian connected* if there exists a hamiltonian path joining any two distinct vertices of G. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. A *hamiltonian cycle* of G is a cycle that traverses every vertex of G. A graph is *hamiltonian* if it has a hamiltonian cycle.

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A *k-container* $C(u, v)$ of G between u and v is a set of k *internally disjoint paths* between u and v. The concept of container is proposed by Hsu [5] to evaluate the performance of communication of an interconnection network. The *connectivity* of G, $\kappa(G)$, is the minimum number of vertices whose removal leaves the remaining graph disconnected or trivial. It follows from Menger's Theorem [11] that there is a k-container between any two distinct vertices of G if G is k-connected.

In this paper, we are interested in specified containers. A k-container $C(u, v)$ of G is a k^*-

container if it contains all vertices of G. A graph G is k^**-connected* if there exists a k^*-container between any two distinct vertices. An 1^*-connected graph is actually a hamiltonian connected graph. Moreover, a 2^*-connected graph with at least three vertices is hamiltonian graph. Thus, the concept of k^*-connected graph is a hybrid concept of connectivity and hamiltonicity. We define the *spanning connectivity* of a graph G, $\kappa^*(G)$, to be the largest integer k such that G is w^*-connected for all $1 \leq w \leq k$ if G is an 1^*-connected graph.

From the application point of view, k^*-containers can be used in multipath communication. Note that graph connectivity and hamiltonicity are two very interesting topics in graph theory. The concept of spanning connectivity is interesting as well. From the theoretical point of view, we should put as much emphasis as we did for hamiltonian graphs. Recently, several families of interconnection networks are proved to be super spanning connected [8, 10, 12]. In this paper, we prove that the $\kappa^*(K(d,t)) = d - 1$ for $d \geq 4$ and $t \geq 1$.

2 The WK-Recursive Networks

The WK-recursive network can be constructed hierarchically by grouping basic modules. A complete graph of any size d can serve as the basic modules. We use $K(d,t)$ to denote a WK-recursive network of level t, each of whose basic modules is a d-vertex complete graph, where $d > 1$ and $t \geq 1$. The structures of $K(7,1)$, $K(7,2)$, and $K(7,3)$ are shown in Figure 1. $K(d,t)$ is defined in terms of a graph as follows:

Each vertex of $K(d,t)$ is labeled as a t-digit radix d number. Vertex $a_{t-1}a_{t-2}\ldots a_1a_0$ is adjacent to (1) $a_{t-1}a_{t-2}\ldots a_1 b$, where $b \neq a_0$ and (2) $a_{t-1}a_{t-2}\ldots a_{j+1}a_{j-1}(a_j)^{j-1}$ if $a_j \neq a_{j-1}$ and $a_{j-1} = a_{j-2} = \ldots = a_0$, where $(a_j)^{j-1}$ denotes $j-1$ consecutive a_js. An *open edge* is incident with $a_{t-1}a_{t-2}\ldots a_0$ if $a_{t-1} = a_{t-2} = \ldots = a_0$. The open edge is reserved for further expansion. Hence, its other end vertex is unspecified. The *open vertex set* O_v of $K(d,t)$ is the set $\{a_{t-1}a_{t-2}\ldots a_0 \mid a_i = a_{i+1} \text{ for } 0 \leq i \leq t-2\}$. In other words, O_v contains those vertices with open edges. Note that $\kappa(K(d,t)) = d - 1$.

Obviously, $K(d,1)$ is a d-vertex complete graph augmented with d open edges. For $t \geq 1$,

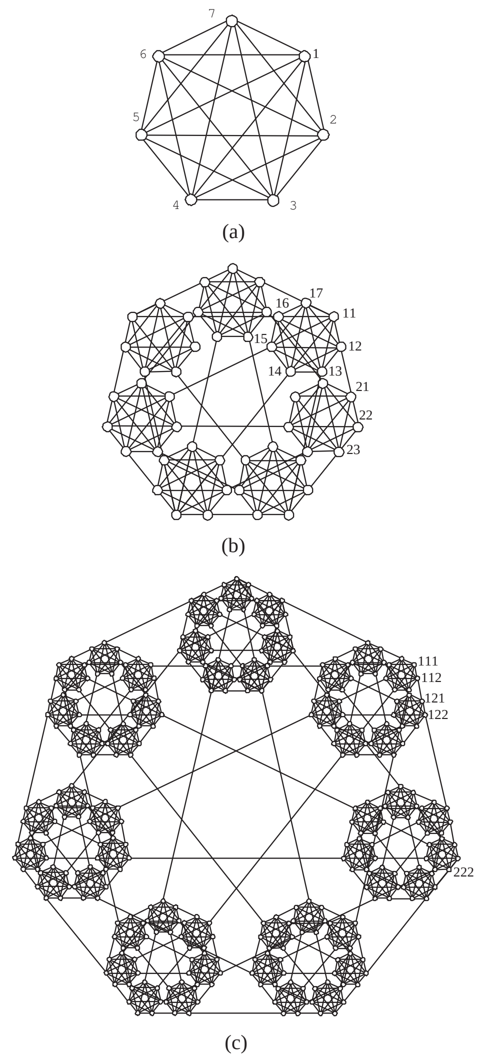

Figure 1: The graphs (a) $K(7,1)$, (b) $K(7,2)$, and (c) $K(7,3)$

$K(d,t+1)$ consists d copies of $K(d,t)$, say $K_1(d,t), K_2(d,t), \ldots, K_d(d,t)$. Thus, we consider $K_i(d,t)$ as the i-th component of $K(d,t+1)$. Let $I = \{w_1, \ldots, w_q\}$ be any q subset of $\{1,2,\ldots,d\}$, we define graph $K_I(d,t)$ is the subgraph of $K(d,t+1)$ induced by $\bigcup_{i=1}^{q} V(K_{w_i}(d,t))$. For $t \geq 2$, the open vertices of $K_i(d,t)$ can be labeled as $o_{i,0}$ and $o_{i,j}$ for $1 \leq i \neq j \leq d$ where $o_{i,0}$ is the only open vertex of $K(d,t+1)$ in $K_i(d,t)$ and $o_{i,j}$ is the vertex in $K_i(d,t)$ joining with the vertex $o_{j,i}$ in $K_j(d,t)$ with an open edge. Note that $(o_{i,j}, o_{j,i})$ is the only edge joining $K_i(d,t)$ to $K_j(d,t)$.

Let $\{r_1, r_2, \ldots, r_q\}$ be a subset of $\{1,2,\ldots,d\}$ for $1 \leq q \leq d$. We define the *ex-*

tended WK-recursive network $\tilde{K}^{r_1,r_2,\ldots,r_q}(d,t)$ as $V(\tilde{K}^{r_1,r_2,\ldots,r_q}(d,t)) = V(K(d,t)) \cup \{w\}$ and $E(\tilde{K}^{r_1,r_2,\ldots,r_q}(d,t)) = E(K(d,t)) \cup \{(o_{j,0},w) \mid j \in \{1,2,\ldots,d\} - \{r_1,r_2,\ldots,r_q\}\}$. For example, $\tilde{K}^1(7,1)$ and $\tilde{K}^{2,3}(7,2)$ are illustrated in Figure 2. Obviously, $\tilde{K}^i(d,t)$ is isomorphic to $\tilde{K}^1(d,t)$ for $1 \leq i \leq d$, $\tilde{K}^{i,j}(d,t)$ is isomorphic to $\tilde{K}^{1,j-i+1}(d,t)$ for $1 \leq i < j \leq d$, and $\delta(\tilde{K}^{r_1,r_2,\ldots,r_q}(d,t)) = d - q$.

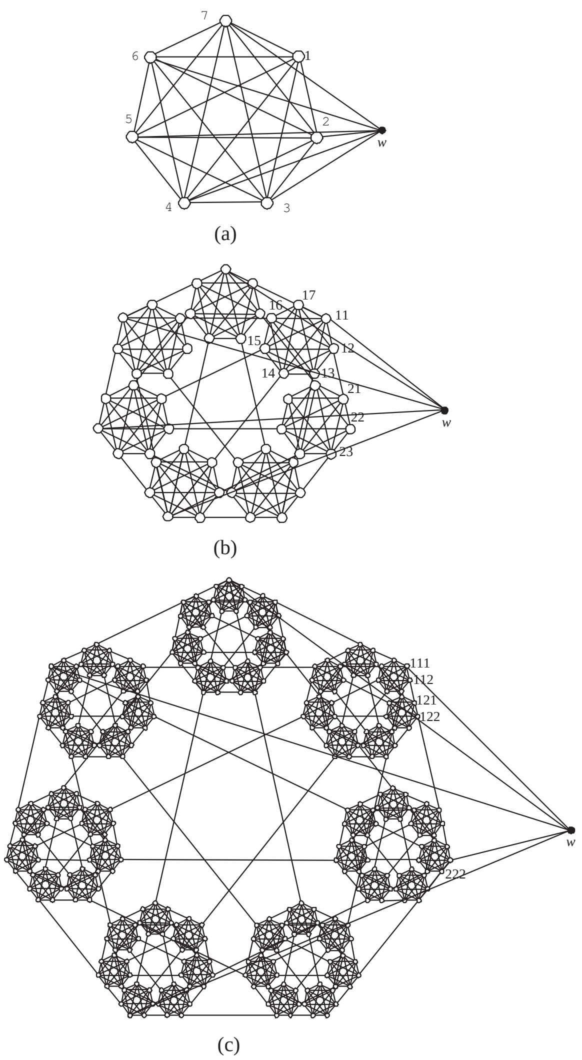

Figure 2: The graphs (a) $\tilde{K}^1(7,1)$ and (b) $\tilde{K}^{2,3}(7,2)$, and $\tilde{K}^{3,5}(7,3)$

Remark 1 *The network $K(2,t)$ is isomorphic to a path with length 2^t, it is easy to check that $K(2,t)$ is not 1^*-connected. By the definition, the spanning connectivity is not defined on $K(2,t)$.*

Remark 2 *We know that $\kappa(K(3,t)) = 2$. Let $\{u,v\}$ be the cut set of $K(3,t)$ for $t \geq 2$, i.e., $K(3,t) - \{u,v\}$ is disconnected. It is impossible to find a hamiltonian path between u and v. Hence, $K(3,t)$ is not 1^*-connected for $t \geq 2$. By the definition, the spanning connectivity is not defined on $K(2,t)$.*

In the following, we consider the WK-recursive network $K(d,t)$ for $d \geq 4$ and $t \geq 1$.

Since $K(d,t)$ is hamiltonian connected for $d \geq 4$ and $t \geq 1$ [4], $K(d,t)$ is 1^*-connected graph. For $d \geq 4$ and $t \geq 1$, $|V(K(d,t))| > 2$, $K(d,t)$ is hamiltonian graph. Hence, $K(d,t)$ is 2^*-connected graph. Therefore, we have the following theorem.

Theorem 1 *$K(d,t)$ is 1^*- and 2^*-connected graph for $d \geq 4$ and $t \geq 1$.*

Lemma 1 *$\kappa^*(\tilde{K}^1(d,t)) = d - 1$ and $\kappa^*(\tilde{K}^{1,j}(d,t)) = d - 2$ for $d \geq 4$ and $t \geq 1$.*

3 Main Result

Theorem 2 *$\kappa^*(K(d,t)) = d - 1$ for $d \geq 4$ and $t \geq 1$.*

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